



**AI/ML-BASED DNS ANOMALY DETECTION WITH INTEGRATED
AND CONTEXTUAL THREAT INTELLIGENCE**

Preliminary Progress Review (PPR) Report

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LIST OF ABBREVIATIONS

Abbreviation	Description
DNS	Domain Name System
ML	Machine Learning
AI	Artificial Intelligence
PPR	Preliminary Progress Review
DGA	Domain Generation Algorithm
C2	Command and Control
DoH	DNS over HTTPS
DoT	DNS over TLS
SOC	Security Operations Center
SIEM	Security Information and Event Management
OSINT	Open Source Intelligence
CVE	Common Vulnerabilities and Exposures
CVSS	Common Vulnerability Scoring System
NVD	National Vulnerability Database
RF	Random Forest
LR	Logistic Regression
IF	Isolation Forest
AE	Autoencoder
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
F1-score	Harmonic Mean of Precision and Recall
PCAP	Packet Capture
DPI	Deep Packet Inspection
SLD	Second-Level Domain
UI	User Interface
API	Application Programming Interface
NLP	Natural Language Processing
NER	Named Entity Recognition

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1. INTRODUCTION

1.1. Purpose

The purpose of this Preliminary Progress Review (PPR) report is to present the current progress and planned direction of the research project on AI/ML-based DNS anomaly detection and integrated threat intelligence. This report provides a structured overview of the research background, problem identification, objectives, methodology, and planned work, enabling an informed assessment of the project's direction and feasibility at this stage.

This PPR is intended to present the conceptual foundation and preliminary research decisions made to date, as well as the planned approach for addressing the identified research problem. It serves as a formal progress checkpoint to ensure alignment with academic expectations, project scope, and research objectives prior to further development and experimentation.

The intended audience of this document includes the project supervisor, academic evaluators, and review panel members involved in the assessment of the Preliminary Progress Review. It may also be referenced by project team members to maintain a shared understanding of the research direction and planned activities.

1.2. Overview

This Preliminary Progress Review report presents the current status and planned direction of the research in a structured manner. The document outlines the background and motivation for the study, followed by an examination of existing research related to DNS-based threats, machine learning-based detection techniques, and DNS-centric threat intelligence.

The study is guided by a clearly defined research problem, supported by specific objectives that outline the intended outcomes. A structured research methodology is proposed, covering the planned approach to data collection, feature engineering, analytical modeling, and evaluation.

In addition, the report outlines the anticipated benefits of the research, the defined scope and expected outcomes, and the constraints that may influence the research process. Supporting materials such as the project plan and schedule are included to demonstrate feasibility and progression planning.

2. LITERATURE REVIEW

2.1. Background and Related Work

The Domain Name System (DNS) is a fundamental component of modern network communication and has increasingly become a target and medium for malicious activities. Due to its essential role, high traffic volume, and historically permissive design, DNS traffic is often trusted and overlooked, making it an attractive channel for attackers. As a result, a wide range of DNS-based threats have emerged, prompting both academic research and industry efforts to improve DNS security.

Recent studies and commercial platforms have explored multiple approaches to detecting malicious DNS activity, including statistical analysis, machine learning-based anomaly detection, and the integration of threat intelligence [1] [2]. While these approaches have shown promise, challenges remain in handling encrypted DNS traffic, adapting to evolving attack techniques, and correlating different categories of DNS abuse in a unified manner [3]. This section reviews relevant background and related work across key areas of DNS-based threat detection.

2.1.1. DNS-Based Threats and Abuse Techniques

DNS-based threats exploit the protocol's role as a fundamental name resolution service to perform malicious activities that closely resemble legitimate traffic. Common abuse techniques include DNS tunneling for covert data exfiltration, DNS-based command-and-control communication, amplification and flooding attacks targeting DNS infrastructure, and the use of dynamically generated domains to evade detection [1] [4].

The increasing adoption of encrypted DNS protocols, such as DNS over HTTPS (DoH) and DNS over TLS (DoT), has further complicated DNS threat detection by limiting direct visibility into query contents [2]. At the same time, attackers continuously evolve their techniques to bypass static detection mechanisms, making traditional signature-based and rule-based approaches insufficient for identifying emerging and previously unseen threats.

Recent research has therefore shifted toward machine learning-based detection, where statistical, lexical, temporal, and behavioral characteristics of DNS traffic are analyzed collectively to identify anomalous or malicious patterns. Rather than relying on individual indicators, ML-based approaches enable the modeling of complex relationships within DNS data, improving adaptability to evolving attack strategies and diverse threat categories.

In addition to technical evasion, DNS-based attacks also exploit limitations in existing threat intelligence ecosystems. Many malicious domains and infrastructures remain active due to delayed intelligence updates, limited DNS-specific context, or overly generic global threat feeds. This allows attackers to reuse infrastructure, rotate domains, or operate within localized environments without timely detection. As a result, DNS abuse often persists even in environments that consume external threat intelligence, highlighting how gaps in intelligence coverage contribute directly to the effectiveness and longevity of DNS-based threats [5].

2.1.2. Machine Learning Approaches for DNS Anomaly Detection & Proactive Threat Intelligence

Machine learning has become a widely adopted approach for DNS anomaly detection due to its ability to model complex patterns within large volumes of DNS data. Unlike traditional signature-based techniques, ML-based approaches can analyze combinations of statistical, lexical, temporal, and contextual features to identify deviations from normal DNS behavior. Prior research has explored supervised, unsupervised, and semi-supervised learning methods to detect both known and previously unseen DNS-based threats [6].

Supervised learning techniques typically rely on labeled datasets to classify DNS traffic as benign or malicious using features such as query length distributions, character entropy, domain structure, query frequency, and timing patterns. While these methods have demonstrated high accuracy under controlled conditions, their effectiveness is often constrained by the availability and quality of labeled data, as well as their ability to generalize to evolving attack techniques. Unsupervised and semi-supervised approaches attempt to address these challenges by modeling normal DNS behavior and identifying anomalies, but may suffer from higher false-positive rates in dynamic environments.

Recent research trends emphasize hybrid ML approaches that combine multiple feature types and analytical perspectives. By jointly considering lexical characteristics, traffic behavior, and temporal correlations, these methods aim to improve robustness against evasion techniques and better handle diverse DNS abuse scenarios [7]. However, many existing studies focus on isolated threat categories or offline evaluation settings, limiting their applicability to real-world, continuously evolving DNS traffic.

In parallel, proactive threat intelligence has been increasingly incorporated into DNS analysis workflows to provide contextual information such as domain reputation, historical activity, and infrastructure associations [8]. When combined with ML-based detection, threat intelligence can assist in interpreting anomalies and prioritizing findings. Nevertheless, threat intelligence sources are often global, delayed, or incomplete, which reduces their effectiveness when used as a standalone mechanism. This limitation highlights the need for detection approaches where machine learning remains the primary analytical driver, with threat intelligence serving as a contextual enhancement rather than a replacement [9].

Overall, existing research demonstrates the potential of machine learning-based DNS anomaly detection while also revealing challenges related to adaptability, data dependency, and contextual awareness. These limitations motivate further investigation into integrated approaches that leverage ML techniques alongside intelligence-driven context to improve detection accuracy and relevance across multiple DNS threat categories.

2.1.3. Detection of DNS Tunneling and Data Exfiltration

DNS tunneling and data exfiltration techniques leverage the DNS protocol to covertly transmit data by embedding encoded information within DNS queries and responses. Because DNS traffic is typically allowed through network boundaries and generated continuously by legitimate applications, tunneling activity can closely resemble normal DNS behavior, making detection particularly challenging [2].

Existing research on DNS tunneling detection has focused on identifying abnormal patterns within DNS traffic, including unusually long domain names, high character entropy, irregular query structures, repetitive request patterns, and atypical query frequencies [10]. These characteristics are widely recognized as effective indicators of tunneling behavior, as they capture deviations from typical DNS usage while remaining observable at the protocol level.

In this research, these DNS traffic characteristics are utilized as feature inputs to machine learning models to support the detection of tunneling and data exfiltration activity. Machine learning–based approaches build upon these indicators by jointly modeling statistical, lexical, and temporal features, allowing relationships among multiple feature types to be learned rather than relying on individual thresholds [11]. This enables more flexible identification of tunneling behavior across varied encoding schemes and transmission strategies.

Despite these advances, several challenges remain in DNS tunneling detection. Many existing approaches assume visibility into plaintext DNS traffic, which is increasingly limited due to the adoption of encrypted DNS protocols [12]. In addition, a significant portion of prior work evaluates detection techniques in offline or controlled experimental environments, where traffic characteristics and attack patterns may not reflect real-world conditions. Tunneling behavior may also be low-rate, intermittent, or intentionally blended with benign traffic, allowing it to evade models trained primarily on high-volume or synthetic datasets. These factors complicate reliable detection in live network environments and increase the likelihood of false positives when legitimate applications generate atypical DNS behavior.

The impact of DNS-based data exfiltration is significant, as it enables prolonged and stealthy leakage of sensitive information while bypassing traditional security controls. Such attacks can remain active over extended periods, particularly when detection mechanisms are unable to operate effectively on live DNS traffic. The persistence of these threats highlights the need for detection techniques that can generalize across diverse tunneling strategies and function under real-time traffic conditions and evolving DNS usage patterns. These challenges motivate continued research into adaptive and context-aware ML-based detection approaches that can identify DNS tunneling activity in near real-time, without relying solely on static indicators or offline analysis.

2.1.4. DGA-Based Domains and DNS-Based C2 Communication

Domain Generation Algorithms (DGAs) are commonly used by malware to generate large numbers of pseudo-random domain names for command-and-control (C2) communication. By frequently changing domains and avoiding static infrastructure, DGAs enable malware to evade traditional blacklist-based defenses. DNS-based C2 communication further complicates detection by blending malicious queries with legitimate DNS traffic, often using low-frequency or encrypted communication patterns.

Research on DGA detection has explored a range of approaches, including lexical analysis of domain names, n-gram modeling, statistical feature extraction, and deep learning–based classification techniques [4] [13]. These methods aim to distinguish algorithmically generated domains from human-readable ones by capturing structural and character-level patterns. In parallel, graph-based and behavioral analysis techniques have been applied to identify relationships between infected hosts, domains, and infrastructure involved in C2 communication [14].

Hybrid detection approaches that combine rule-based filtering with machine learning models have demonstrated improved effectiveness, particularly in identifying previously unseen DGAs [15]. By using simple heuristics to eliminate obvious benign domains and applying ML-based analysis to more ambiguous cases, such approaches can reduce computational overhead while maintaining detection accuracy. However, many existing studies treat DGA detection and C2 communication analysis as separate problems, limiting their ability to correlate malicious activity across different stages of the attack lifecycle [16].

Several challenges remain in detecting DGA-based domains and DNS-based C2 traffic in operational environments. A significant portion of prior work relies on offline datasets and controlled experiments, which may not accurately represent real-world DNS traffic dynamics. Additionally, modern malware increasingly uses encrypted DNS protocols, low-and-slow communication patterns, and infrastructure reuse strategies that complicate real-time detection and correlation.

The impact of undetected DGA-based C2 communication is substantial, as it enables persistent control over compromised systems and facilitates further malicious activity. These challenges highlight the need for adaptive ML-based detection techniques that can operate on live DNS traffic, correlate DGA activity with C2 behavior, and generalize across evolving malware strategies without relying solely on static indicators.

2.1.5. Detection of DNS Abuse and Infrastructure Attacks

DNS abuse and infrastructure attacks target the availability, reliability, and performance of DNS services rather than focusing solely on data exfiltration or covert communication. Common forms of abuse include DNS amplification attacks, flooding attacks, reflection-based denial-of-service activity, and misuse of DNS infrastructure for large-scale malicious campaigns. These attacks often generate high volumes of DNS traffic that can disrupt normal resolution services and impact dependent applications.

Prior research in this area has largely focused on traffic-level analysis, including measurements of query rates, response sizes, traffic asymmetry, and temporal burst patterns [1]. Both statistical techniques and machine learning-based approaches have been explored to identify deviations from expected DNS behavior. Supervised and semi-supervised models have been applied using features derived from traffic intensity, packet distributions, and temporal correlations to distinguish abusive activity from legitimate high-volume DNS usage [17].

However, detecting DNS abuse at the infrastructure level presents several challenges. Many existing approaches are evaluated in offline or partially simulated environments, limiting their ability to reflect real-world DNS traffic variability and scale. In operational networks, DNS traffic patterns can change rapidly due to legitimate events such as flash crowds, software updates, or misconfigurations, increasing the difficulty of distinguishing abuse from benign anomalies [18]. These factors contribute to elevated false-positive rates when detection models are not sufficiently adaptive.

The impact of DNS abuse and infrastructure attacks is significant, as sustained or large-scale attacks can degrade service availability and affect downstream network operations. The need to analyze high-throughput DNS traffic in near real-time further complicates detection, particularly in large or distributed environments. These challenges motivate continued research into scalable, adaptive ML-based detection techniques that can operate effectively on live DNS traffic while maintaining robustness against evolving abuse strategies.

2.1.6. Contextual Threat Intelligence Platforms

Threat intelligence platforms aggregate information from multiple sources, including open-source intelligence (OSINT), reputation databases, historical incident reports, and observed threat activity, to provide contextual insights into malicious behavior. When applied to threat analysis, threat intelligence can support the identification of malicious domains, infrastructure reuse, and relationships between observed system activity and known threat actors or campaigns.

Existing threat intelligence solutions are typically designed to provide broad, global coverage across multiple attack vectors. While this enables large-scale visibility, such intelligence is often generic in

nature and may lack contextual or relevance to localized environments and sector specific environments [5]. As a result, threats that rely on short-lived domains, infrastructure rotation, or region-specific activity can remain undetected or be identified only after significant delay.

Research has explored intelligence approaches that emphasize contextual correlation, behavioral patterns, and infrastructure-level relationships rather than static reputation alone. These approaches aim to enhance analytical understanding by linking anomalies with intelligence derived from historical observations, related domains, and auxiliary data sources [8]. However, many existing intelligence-driven methods operate independently of real-time detection systems or rely on post-event analysis, limiting their effectiveness in dynamic operational settings.

Another challenge lies in the integration of threat intelligence with machine learning–based detection. Intelligence feeds may contain incomplete, noisy, or outdated information, which can reduce reliability if treated as a primary detection mechanism. Consequently, threat intelligence is most effective when used to enrich and contextualize ML-based detections rather than replace them. This highlights the importance of intelligence frameworks that are adaptable, transparent, and capable of incorporating analyst input to refine relevance and prioritization.

Overall, existing work demonstrates the value of threat intelligence in enhancing anomaly detection while also revealing limitations related to timeliness, specificity, and integration [9]. These observations motivate further research into intelligence frameworks that complement ML-based detection, support proactive analysis, and improve the interpretability and relevance of detected threats across diverse environments.

2.2. Identification and Significance of the Problem

The Domain Name System (DNS) is a fundamental component of modern network communication and is often implicitly trusted within organizational environments. Due to its essential role and high volume of legitimate traffic, DNS activity frequently receives less security analysis compared to other network protocols. This makes DNS an attractive and effective vector for attackers to conduct stealthy and persistent malicious activities [1].

An analysis of existing research and industry practices reveals several key problems associated with DNS-based threat detection:

- DNS is commonly under-monitored, allowing malicious activity to blend with normal name resolution traffic.
- Attackers exploit DNS for multiple purposes, including data exfiltration, command-and-control communication, and large-scale infrastructure abuse.
- Many existing detection mechanisms rely on static rules, reputation lists, or signature-based techniques that struggle to identify novel or evolving DNS threats.
- A significant portion of ML-based DNS detection research is evaluated in offline or controlled environments, limiting its applicability to real-world, high-volume DNS traffic.
- The increasing adoption of encrypted DNS protocols reduces visibility into DNS query content, further challenging detection approaches that depend on direct inspection [12].
- Existing enterprise security monitoring platforms typically ingest DNS alerts as static events, offering limited flexibility to incorporate environment-specific context or analyst-driven insights.

- Most current systems do not allow detections to be augmented with human intelligence, such as contextual allowlisting or organization-specific knowledge, due to restricted configurability of underlying detection mechanisms.
- Threat intelligence platforms often provide generalized insights that lack localization, sector specification and effective correlation with live detection outputs, limiting their ability to anticipate emerging or environment-specific threats [5].

The significance of these problems is substantial. DNS-based attacks can remain undetected for extended periods, enabling prolonged data leakage, persistent malware control, and disruption of critical services. The absence of adaptive, real-time detection and contextual enrichment limits situational awareness and reduces the operational usefulness of DNS security monitoring. These challenges highlight the need for research into ML-driven DNS anomaly detection approaches that can operate on live traffic, generalize across diverse DNS threat categories, and be enriched with contextual intelligence and analyst input to improve relevance, transparency, and practical value in enterprise environments.

3. RESEARCH QUESTION

This research aims to investigate the effectiveness of machine learning-based approaches for detecting malicious DNS activity in modern network environments.

The primary research question of this study is:

“How can machine learning-based DNS anomaly detection be designed to identify diverse DNS-based threats in real-time environments while incorporating contextual threat intelligence and human knowledge, and enabling centralized monitoring within enterprise security platforms?”

To support this investigation, the following sub-questions are considered:

- How can ML-based techniques be applied to detect multiple DNS threat categories, including DNS tunneling, infrastructure abuse, DGA-based domains, and DNS-based C2 communication?
- How can contextual threat intelligence be combined with ML-based detection to enhance the interpretation and prioritization of DNS anomalies?
- How can analyst-provided human intelligence be incorporated to refine detection outputs in enterprise monitoring environments?
- How can detected DNS anomalies be integrated into centralized enterprise security monitoring platforms to support unified visibility and analysis?

4. RESEARCH OBJECTIVES

4.1. Main Research Objective

The main objective of this research is to design and evaluate a machine learning-based DNS anomaly detection approach capable of detecting diverse DNS-based threats in real-time environments, while incorporating contextual threat intelligence and human knowledge to support centralized monitoring in enterprise security platforms.

4.2. Specific Research Objectives

- Analyze DNS traffic characteristics relevant to detecting malicious activities including DNS tunneling, infrastructure abuse, DGA-based domains, and DNS-based command-and-control communication.
- Identify and extract suitable statistical, lexical, temporal, and contextual features for ML-based DNS anomaly detection under real-world traffic conditions.
- Investigate machine learning techniques for detecting multiple DNS threat categories in near real-time environments.
- Explore methods for integrating contextual threat intelligence to enhance the interpretation and prioritization of detected DNS anomalies.
- Incorporate analyst-driven human intelligence to refine detection outputs and improve adaptability to environment-specific contexts.
- Enable integration of detection results into centralized enterprise security monitoring platforms to support unified visibility and analysis.

5. RESEARCH METHODOLOGY

5.1. Overall Research Approach

This research adopts a design-oriented and experimental research approach to investigate the application of machine learning techniques for detecting anomalous and malicious DNS activity in modern network environments. The study is structured around the analysis of DNS traffic data, the extraction of relevant features, and the evaluation of ML-based detection techniques across multiple DNS threat categories.

The research follows a data-driven methodology, where DNS traffic data is collected from appropriate sources and analyzed to identify patterns associated with both benign and malicious behavior. Based on insights from the literature, suitable features are derived to represent statistical, lexical, temporal, and contextual characteristics of DNS activity. These features form the basis for training and evaluating machine learning models designed to detect anomalies and threat-specific behaviors.

An iterative experimental process is employed, allowing models and feature sets to be refined based on observed performance and analytical findings. Detection effectiveness is evaluated under conditions that reflect real-world DNS usage, with emphasis on adaptability, generalization, and near real-time applicability. The approach also considers the integration of contextual threat intelligence and analyst-provided inputs as supplementary information to enhance detection relevance and interpretability.

Overall, the research approach combines experiential evaluation with analytical modeling, guided by established research in DNS security and machine learning, to address the identified research problem in a systematic and academically grounded manner.

5.2. Conceptual Framework and System Overview

The conceptual framework of this research is centered on a DNS-centric anomaly detection pipeline that integrates machine learning–based analysis with contextual threat intelligence and analyst-provided inputs. The framework is designed to support the detection of multiple DNS-based threat categories while enabling centralized visibility within enterprise monitoring environments.

At a high level, the framework begins with the ingestion of DNS traffic data, which is processed to extract relevant characteristics representing normal and anomalous behavior. These characteristics are transformed into structured feature representations capturing statistical, lexical, temporal, and contextual aspects of DNS activity. The extracted features serve as inputs to machine learning–based detection components responsible for identifying anomalous or suspicious patterns across different DNS threat categories.

The framework incorporates multiple analytical components, each aligned with a specific class of DNS threats, including DNS tunneling and data exfiltration, DNS abuse and infrastructure attacks, and DGA-based domains with DNS-based command-and-control communication. Outputs from these components are analyzed and correlated to provide a unified view of detected DNS anomalies.

Contextual threat intelligence is incorporated into the framework as a customizable and proactive intelligence layer that operates alongside machine learning–based detection. This intelligence layer supports early contextual awareness, assists in enabling environment-specific contextualization of analytical outputs. Analyst-provided human intelligence can be incorporated to adapt and refine this intelligence over time, improving relevance and analytical value while complementing ML-based detection.

Finally, the framework supports the export of detection results to enterprise security monitoring platforms, enabling centralized visibility and analysis. This conceptual design emphasizes modularity, adaptability, and extensibility, allowing the research framework to evolve as new DNS threat patterns and analytical techniques emerge.

5.3. Data Collection and Data Sources

The research utilizes DNS-related data obtained from multiple sources to support the analysis of both benign and malicious DNS behavior. These data sources are selected to enable effective feature extraction, model training, and evaluation across multiple DNS-based threat scenarios.

Publicly available datasets are leveraged to represent known benign and malicious DNS behavior, including datasets that capture DNS abuse patterns, DGA-generated domains, and infrastructure-level attacks. These datasets provide labeled reference data that supports supervised and semi-supervised learning approaches. In addition, DNS traffic is obtained through controlled network monitoring to observe realistic DNS query behavior and extract relevant traffic characteristics for analysis.

To address gaps in existing datasets and improve coverage of specific attack behaviors, supplementary data is generated using appropriate tools and techniques. This approach allows the research to model control cases, evolving attack patterns, and low-frequency behaviors that may not be sufficiently represented in public datasets alone. The combination of collected, captured, and generated data supports a more comprehensive representation of DNS activity.

All collected data undergoes preprocessing steps such as cleaning, normalization, labeling, and feature extraction prior to use in analysis. The processed datasets are used for model training, validation, and selected real-time evaluation experiments, depending on the requirements of each detection component. Together, these data sources enable consistent experimentation while maintaining flexibility to accommodate different DNS threat categories and research objectives

5.4. Feature Engineering and Analysis

Feature engineering plays a central role in enabling effective detection of DNS-based threats within the proposed research. The system derives meaningful features from raw DNS traffic and related metadata to capture characteristics associated with both benign behavior and diverse malicious activities.

The engineered features are designed to represent multiple analytical perspectives, including lexical properties, statistical distributions, temporal patterns, and behavioral characteristics observable within DNS traffic. Examples include domain name structure attributes, query length and entropy-related measures, frequency and timing patterns, request–response ratios, and variations across query sequences. All features used for model training are selected such that they can be extracted from live DNS traffic streams, ensuring consistency between offline analysis and real-time deployment.

Feature analysis is conducted to assess relevance, stability, and discriminative power across different threat categories. Redundant or low-impact features are reduced through selection techniques to improve model efficiency and generalization.

In addition to automated feature extraction, the framework supports the incorporation of human-informed contextual inputs through a rule-based mechanism. This enables environment-specific adjustments, such as contextual exclusions or prioritization logic, which complement the learned feature representations and enhance analytical relevance without altering core detection models.

Overall, the feature engineering process is structured to balance detection accuracy, interpretability, scalability, and a strong emphasis on recall, ensuring that anomalous and potentially malicious DNS activity is identified with minimal risk of missed detections.

5.5. Detection Models and Analytical Techniques

5.5.1. DNS Tunneling and Data Exfiltration Detection

Table 1: Plaintext DNS Tunneling Detection

Plaintext DNS Tunneling Detection	
Detection Objective	
Primary Detection Goal	Detect abnormal DNS queries that indicate potential DNS tunneling or data exfiltration by evaluating query characteristics and applying event-level aggregation across related DNS activity in plaintext traffic.
Motivation	DNS tunneling enables stealthy data leakage through a protocol that is usually allowed in networks, missed detection can result in long-duration exposure and loss of sensitive data.
Input Data and Features	
DNS Data Used	DNS query names and query-level metadata that can be observed from live DNS traffic (sufficient to compute lexical + flow/statistical features).
Feature Types	Lexical - query string characteristics such as length/structure/entropy-style indicators Behavioural/Statistical - flow-level patterns, distributions, query behaviour
Real-Time Feasibility	Features are chosen to be extracted from live plaintext DNS traffic, aggregation is performed over observed events/SLD groupings.

Analytical / Model-Based Approach	
Detection Approach	Fusion-based ML + Event aggregation Fusion - combines probability outputs from lexical and behavioural detection models into a single risk score Event aggregation - groups related DNS queries over time to elevate sustained suspicious activity to an event-level detection
Model Type(s)	Lexical model - Logistic Regression using lexical DNS features (probability output) Behavioural model - Logistic Regression using behavioural and statistical DNS features (probability output) Fusion - Weighted fusion combining probability outputs from the lexical and behavioural models using predefined weights, followed by rule-driven hybrid logic to refine the final detection decision
Rationale	Lexical features capture payload-like strings & behavioural features capture traffic behaviour that bypasses purely lexical signals. Fusion reduces single-model weaknesses & aggregation improves reliability in real-time streams by reducing “one-off” noise.
Detection Output and Scoring	
Output Type	Lexical Model Output - Produces a probability score per DNS query based on lexical characteristics indicative of anomalous behavior. Behavioural Model Output - Produces a probability score per DNS query based on behavioural and statistical traffic patterns. Fusion Output - Combines lexical and behavioural probability scores into a single risk score using weighted fusion with rule-driven logic. Event Aggregation Output - Aggregates related DNS queries over time to generate an event-level score and label for analyst review.
Severity Mapping	Severity is derived from fused probability/ event risk level (OK / SUSPICIOUS / ALERT) and then mapped to final alert severity.
Context Provided	Domain and SLD grouping, timestamps, involved host identifiers, and contributing model scores, providing contextual information to support analyst understanding of what was detected, why it was flagged, where it originated, who was involved, and when the activity occurred.
Evaluation Focus	
Primary Evaluation Metric	Focus on recall and accuracy, with priority given to recall to minimize missed true positives while maintaining balanced accuracy.
Reason for Metric Choice	Exfiltration is high-impact and false negatives are more damaging than manageable false positives, especially when alerts are reviewed in a SOC/SIEM flow.
Known Limitations	Tunneling activity that closely resembles benign DNS patterns and behaviors can be difficult to distinguish. Enterprise DNS usage varies across environments, and aggregation rules and thresholds may require tuning to control false positives.

Table 2: DoH Tunneling Detection

DoH Tunneling Detection	
Detection Objective	
Primary Detection Goal	Detect DNS tunneling / exfiltration behaviour occurring over DoH using observable DoH flow/statistical characteristics and multi-model fusion.
Motivation	DoH reduces visibility into plaintext query contents, and therefore detection must rely only on traffic characteristics to identify suspicious patterns without payload inspection.
Input Data and Features	
DNS Data Used	Encrypted DNS-over-HTTPS (DoH) traffic metadata and flow-level measurements, including packet-level statistics, flow volume characteristics, timing and rate-based information, and session-level behavioural attributes observable without access to DNS query payloads.
Feature Types	Statistical and behavioural features including traffic volume metrics, packet size patterns, timing and rate-based characteristics, and session-level behaviour, selected for real-time extraction and tested using minimal feature configurations.
Real-Time Feasibility	Only features that can be extracted within real-time limits are used & models were evaluated in real-time context with reduced feature sets.
Analytical / Model-Based Approach	
Detection Approach	Fusion-based, multi-detector ensemble combining supervised and unsupervised detectors, followed by event-level aggregation logic.
Model Type(s)	Random Forest (RF) - Acts as the primary decision-making classifier Logistic Regression (LR) - Provides a lightweight probabilistic classifier Isolation Forest (IF) - Is trained primarily on benign DoH traffic to model normal behavior and detect anomalous deviations Autoencoder (AE) - Is trained to learn characteristic representations of malicious DoH traffic patterns and identifies deviations Fusion - Combines outputs from RF, LR, IF, and AE at the score and label level to leverage both supervised classification and anomaly detection signals
Rationale	DoH detection benefits from multiple viewpoints: RF/LR capture discriminative patterns, AE/IF capture deviations and rare behaviours, fusion reduces over-reliance on one detector and helps maintain recall under varied traffic.
Detection Output and Scoring	
Output Type	RF Output - Produces a classification score and label indicating the likelihood of malicious DoH activity LR Output - Produces a probabilistic score representing the likelihood of malicious behavior IF Output - Produces an anomaly score indicating how much the observed DoH traffic deviates from learned benign behavior AE Output - Produces a reconstruction error score that reflects similarity to learned malicious traffic patterns Fusion Output - Combines RF, LR, IF, and AE outputs into a unified risk score or decision using score- and label-level fusion with rule-based refinement Event Aggregation Output - Aggregates related DoH detections over time to generate an event-level score and severity label suitable for alerting and analyst investigation.
Severity Mapping	Severity derived from fused outcome (high-confidence - malicious vs low-confidence - suspicious) and can be mapped into SIEM-friendly alert severities based on fused score thresholds / rule conditions.

Context Provided	Flow identifiers, timestamps, source host information, and per-detector outputs (RF, LR, IF, AE), providing sufficient context to understand what activity was detected, why it was flagged, who initiated it, when it occurred, and where it originated.
Evaluation Focus	
Primary Evaluation Metric	Focus on recall and accuracy, with priority given to recall to minimize missed true positives while maintaining balanced accuracy.
Reason for Metric Choice	DoH threats are harder to inspect, prioritising recall reduces the chance of silent exfiltration over encrypted channels.
Known Limitations	Real-time DoH feature limits reduces model richness Fusion rules/thresholds require tuning for different network baselines Drift in DoH usage patterns may impact stability over time

5.5.2. DGA-Based DNS and C2 Communication Detection

Table 3: DGA-Based DNS Detection Model

DGA-Based DNS Detection Model	
Detection Objective	
Primary Detection Goal	Detect algorithmically generated domain (DGA) queries in DNS traffic that are commonly used by malware to establish Command-and-Control (C2) communication channels.
Motivation	DGAs enable malware to evade static blacklists by continuously generating new domains. Since DNS is often the first observable communication step in malware C2 activity, detecting DGA-based domains provides an early and reliable indicator of compromise.
Input Data and Features	
DNS Data Used	DNS query domain names extracted from live or recorded DNS traffic. The detection focuses on queried domain strings without relying on payload inspection or encrypted traffic decryption.
Feature Types	Lexical features derived from domain names, including character-level n-grams, structural patterns, and statistical properties implicitly learned by the model.
Real-Time Feasibility	Yes. All features are extracted directly from DNS queries and can be computed in real time with minimal processing overhead.
Analytical / Model-Based Approach	
Detection Approach	Machine-learning-based detection with comparative evaluation of multiple models. The final design prioritises probabilistic scoring to support flexible operational thresholds.
Model Type(s)	Logistic Regression with character n-gram features (primary model), XGBoost (tree-based baseline), and character-level Convolutional Neural Networks for comparison.
Rationale	Character-level lexical analysis is well-suited for DGA detection as it captures patterns produced by domain generation algorithms. Logistic Regression offers strong generalisation, interpretability, and low inference latency, making it suitable for real-time DNS monitoring environments.
Detection Output and Scoring	

Output Type	For each DNS query, the system outputs a probability score ($p \in [0,1]$) indicating the likelihood that the queried domain is algorithmically generated (DGA-based). Based on this score, the query is classified as Benign or DGA using a configurable decision threshold.
Severity Mapping	The probability score is mapped to operational severity levels. Low scores are logged as OK (benign activity), while higher scores trigger Warning or Alert events, indicating increased confidence of DGA-based malicious behaviour.
Context Provided	Each detection record includes the timestamp, queried domain name, model confidence score, and classification outcome. This contextual information supports analyst interpretation and enables integration with SIEM and security monitoring workflows.
Evaluation Focus	
Primary Evaluation Metric	ROC-AUC, precision, recall, and confusion matrix analysis are used to evaluate detection performance across different operational scenarios and decision thresholds.
Reason for Metric Choice	ROC-AUC measures overall discrimination capability independent of threshold selection, while precision and recall capture the trade-off between false positives and missed DGA detections, which is critical for DNS-based security monitoring.
Known Limitations	Wordlist-based DGAs may resemble legitimate domains and are harder to distinguish using lexical features alone. Detection performance is also influenced by threshold selection, requiring careful tuning to balance recall and false positives in operational environments.

Table 4: Behavioural Rule Engine for DGA and C2 Detection

Behavioural Rule Engine for DGA and C2 Detection	
Detection Objective	
Primary Detection Goal	Detect anomalous DNS query behaviour at the host level that may indicate automated DGA activity and early-stage DNS-based C2 communication.
Motivation	Malware C2 communication often exhibits behavioural patterns such as query bursts and high rates of unique domain lookups. These behaviours provide complementary indicators beyond lexical domain analysis.
Input Data and Features	
DNS Data Used	DNS query metadata extracted from live or recorded DNS traffic, including source host, queried domain name, and timestamp.
Feature Types	Behavioural and temporal features derived from DNS activity, such as query rate per host and number of unique domains queried over time.
Real-Time Feasibility	Yes. All features are computed incrementally from streaming DNS metadata with low processing overhead, enabling real-time monitoring.
Analytical / Model-Based Approach	
Detection Approach	Rule-based behavioural analysis applied over sliding time windows to identify anomalous DNS usage patterns.

Rule Type(s)	Query burst detection and high unique-domain-rate detection per host.
Rationale	DGA-driven malware frequently probes multiple domains in rapid succession to locate an active C2 server. These patterns are difficult to detect using per-domain lexical features alone.
Detection Output and Scoring	
Output Type	Behavioural severity classification (OK / SUSPICIOUS) with rule-based explanations indicating the detected anomaly.
Severity Mapping	Behavioural rule triggers generate warning-level events, which are later combined with ML-based DGA probability scores in the decision layer.
Context Provided	Each detection includes host identifier, observation window, query counts, unique domain counts, and triggered rule indicators.
Evaluation Focus	
Primary Evaluation Metric	Rule trigger accuracy, false alert rate, and consistency of behavioural detection under realistic DNS traffic conditions.
Reason for Metric Choice	Behavioural rules are intended to provide early warning signals rather than final classification; therefore, stability, explainability, and low false positives under benign traffic are prioritized.
Known Limitations	High-volume legitimate DNS activity may trigger behavioural rules. As a result, rule outputs must be interpreted alongside ML-based domain classification to avoid false positives.

5.5.3. DNS Abuse and Infrastructure Attack Detection

Table 5: NON-DOH/ DOT Infrastructure Attack Detection

NON-DOH/ DOT Infrastructure Attack Detection	
Detection Objective	
Primary Detection Goal	Identify DNS abuse and infrastructure attacks including Amplification, Reflection, Water Torture, Query Flood, Cache Poisoning, and TCP SYN Flood targeting traditional DNS infrastructure (UDP/TCP port 53)
Motivation	DNS infrastructure attacks can cause service disruption, bandwidth exhaustion, and server resource depletion. Early detection prevents DDoS impact, protects DNS availability, and maintains network integrity for critical name resolution services.
Input Data and Features	
DNS Data Used	Network flow data from live capture (Pcap4J) or PCAP files, captures bidirectional DNS traffic on UDP/TCP port 53 with full packet payload access for Deep Packet Inspection (DPI)
Feature Types	41 engineered features across 6 categories (1) Network Identity (5-tuple), (2) Flow Statistics (duration, packet/byte counts), (3) Timing Features (IAT mean/std/max), (4) Size Features (packet length mean/std/max), (5) DNS Header Features (QR, Opcode, QDCOUNT, query types,

	answer counts), (6) DNS Volume/Derived Features (amplification factor, query/response ratio, queries per second, ANY/TXT ratios, EDNS parameters)
Real-Time Feasibility	High feasibility - Flow-based aggregation with 120-second timeout enables near real-time detection; feature extraction adds <100ms latency; XGBoost inference completes in < 10ms per flow, end-to-end detection achievable within 1-2 seconds of attack initiation
Analytical / Model-Based Approach	
Detection Approach	Supervised machine learning with hybrid rule-based validation , binary classification (BENIGN vs ATTACK) followed by attack type attribution using feature patterns
Model Type(s)	XGBoost Gradient Boosting Classifier: ensemble of decision trees with gradient boosting optimization
Rationale	XGBoost selected for: (1) High accuracy (95%+ on test data), (2) Handles imbalanced datasets effectively, (3) Built-in feature importance for explainability, (4) Fast inference for real-time deployment, (5) Robust to missing values and outliers, (6) Well-suited for tabular network flow data
Detection Output and Scoring	
Output Type	Binary classification label (0=BENIGN, 1=ATTACK) with probability scores for each class, includes attack type attribution (Amplification, Reflection, Water Torture, Query Flood, Cache Poisoning, TCP SYN Flood) based on feature signatures
Severity Mapping	Critical (prob ≥ 0.9): High-confidence attacks with immediate impact High (prob 0.7-0.89): Probable attacks requiring investigation Medium (prob 0.5-0.69): Suspicious patterns needing validation Low (prob 0.3-0.49): Anomalous behavior flagged for monitoring
Context Provided	Feature importance scores, confidence level, attack type classification, source/destination IPs and ports, attack signature (e.g., "High amplification factor: 15.2x", "Queries per second: 1,523"), temporal context (flow duration, packet rates), comparison to baseline behavior
Evaluation Focus	
Primary Evaluation Metric	F1-Score (harmonic mean of precision and recall) with target ≥ 0.90 ; secondary metrics include Accuracy, ROC-AUC, Precision (attack class), and Recall (attack class)
Reason for Metric Choice	F1-Score chosen because: (1) Balances false positives (benign flagged as attack) and false negatives (missed attacks), (2) Critical in security applications where both types of errors have consequences, (3) Accounts for class imbalance in attack/benign distribution, (4) Single metric for model comparison across retraining iterations, (5) Interpretable for stakeholders (90% F1 = strong balanced performance)

Known Limitations	(1) Zero-day attacks: Novel attack patterns not in training data may evade detection, (2) Feature drift: Network behavior changes reduce accuracy over time without retraining, (3) Encrypted payload attacks: Cannot detect attacks hidden within encrypted DNS extensions, (4) Low-and-slow attacks: Stealthy attacks below statistical thresholds may bypass detection, (5) Legitimate high-volume: CDN DNS or popular domains may trigger false positives, (6) Adversarial evasion: Attackers may craft packets to mimic benign patterns
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5.5.4. Threat Intelligence Correlation and Scoring

Table 6: Target Location Detection (Country Classifier)

Target Location Detection (Country Classifier)	
Detection Objective	
Primary Detection Goal	Identify the specific Victim Country mentioned in a cyber-attack news article
Motivation	Threat intelligence is useless without context. Knowing who is being attacked helps organizations filter threats relevant to them.
Input Data and Features	
DNS Data Used	Custom Manually Labeled Dataset - A curated collection of cybersecurity news articles, labeled using a LLM with the correct victim nation.
Feature Types	Contextual Embeddings - The model utilizes DistilBERT embeddings to understand sentence structure (e.g., "attack targeting Japan" vs. "hackers from Japan").
Analytical / Model-Based Approach	
Detection Approach	Supervised Fine-Tuning - We applied Transfer Learning to a pre-trained distilbert-base-uncased model.
Rationale	Standard NER (Named Entity Recognition) models often confuse the attacker's location with the victim's. A custom model is required to understand the context of "victimhood."
Detection Output and Scoring	
Output Type	Country Label (e.g., "France", "United States")
Severity Mapping	Location itself does not determine severity. However, the Confidence Score is used as a quality gate. Confidence < 0.50: Result is flagged as "Unsure" or discarded to prevent false geolocation.
Evaluation Focus	
Primary Evaluation Metric	Accuracy & F1-Score - F1-score is critical to balance precision
Reason for Metric Choice	F1-Score was chosen to ensure the model correctly identifies both frequent and rare victim nations in imbalanced dataset, effectively balancing Precision and Recall.
Known Limitations	May struggle with implied locations (e.g., mentioning "Paris" implies "France") unless explicitly trained on those mappings.

Table 7: Industry Sector Classification

Industry Sector Classification	
Detection Objective	
Primary Detection Goal	Classify the victim organization into a standard industrial sector (e.g., Financial Services, Healthcare, Energy).
Motivation	Different sectors face different types of threats. Filtering by sector allows organizations to focus only on threats relevant to their industry.
Input Data and Features	
DNS Data Used	VCDB (Veris Community Database) - A publicly available, open-source dataset of historical security incidents, pre-labeled with industry sectors.
Feature Types	Semantic Text Features - The model learns to associate specific terms (e.g., "patient records", "SCADA") with their respective sectors.
Analytical / Model-Based Approach	
Detection Approach	Supervised Fine-Tuning - Similar transfer learning approach as Component 1, but applied to a multi-class classification problem (5-7 distinct sectors).
Rationale	Using VCDB provided a standardized, industry-accepted taxonomy for sectors, avoiding ambiguous labels.
Detection Output and Scoring	
Output Type	Sector Label (e.g., "Financial Services", "Healthcare", "Technology")
Severity Mapping	Critical Infrastructure (Financial, Healthcare): Acts as a multiplier, increasing the final triage priority.
Evaluation Focus	
Primary Evaluation Metric	Macro-Average F1 Score - Essential because the dataset was imbalanced
Reason for Metric Choice	Need to ensure the model performed well on rare sectors and didn't just bias toward the most common ones.
Known Limitations	Because of the imbalance of the dataset accuracy of some sectors are low.

Table 8: Threat Scoring & Triage Engine

Threat Scoring & Triage Engine	
Detection Objective	
Primary Detection Goal	Assign a precise Severity Score (0.0 - 10.0) and Triage Label (Low, Medium, High, Critical) to every analyzed article.
Motivation	To automate the prioritization process, ensuring analysts focus on active exploitations and critical vulnerabilities rather than informational noise.
Input Data and Features	
DNS Data Used	Extracted text, Regex-matched CVE patterns (e.g., CVE-2024-XXXX), and NVD API responses.
Feature Types	Hybrid Features 1. External Intelligence - Official CVSS scores from the NIST National Vulnerability Database. 2. Lexical Indicators - Weighted keywords indicating urgency (e.g., "Active Exploitation," "PoC Released").
Analytical / Model-Based Approach	

Detection Approach	Hybrid Cascading Logic (Deterministic + Heuristic) - A two-stage waterfall approach. Conditional Logic Engine: • Stage 1 (Priority): Regex scan for CVE IDs. If found → Query NVD API for the official CVSS score. • Stage 2 (Fallback): If no CVE is found → Execute Rule-Based Heuristics (summing weights of high-risk keywords) to estimate severity.
Rationale	This hybrid approach maximizes accuracy. We rely on official government data (NVD) whenever possible for precision, but fall back to our custom rules to catch "Zero-Day" attacks that don't have a CVE ID yet.
Detection Output and Scoring	
Output Type	Risk Score (Float 0-10) and Categorical Label.
Severity Mapping	Standard CVSS Mapping: • 9.0 - 10.0: Critical • 7.0 - 8.9: High • 4.0 - 6.9: Medium • 0.0 - 3.9: Low
Context Provided	The specific logic trigger (e.g., "Score based on CVE-2023-1234" OR "Score based on keyword 'Ransomware'").
Evaluation Focus	
Primary Evaluation Metric	Logic Traceability & Consistency - Verifying that specific inputs always result in the expected score (Deterministic Testing).
Reason for Metric Choice	Unlike probabilistic AI models, a scoring engine requires 100% reproducibility. A "Critical" vulnerability must never be randomly scored as "Low."
Known Limitations	Dependency on external NVD API availability; potential for "keyword stuffing" in sensationalist news articles causing temporarily inflated scores.

5.6. Data Analysis and Evaluation Methods

The performance of the proposed DNS anomaly detection models is evaluated using standard classification metrics, with emphasis on their suitability for security-critical environments.

The primary evaluation metrics used in this research are:

- Recall, to measure the ability of the system to correctly identify malicious DNS activity
- Accuracy, to assess overall correctness of predictions
- Precision, to evaluate the proportion of correct positive detections
- F1-score, to balance recall and precision in performance assessment

Recall is given higher priority during evaluation, as missing a true positive (i.e., failing to detect malicious DNS activity) poses a significantly greater risk than generating manageable false positives. This is particularly important for DNS tunneling, data exfiltration, and encrypted DNS abuse, where undetected activity can persist over long periods.

Model performance is further analyzed by:

- Evaluating different decision thresholds and configurations

- Monitoring the trade-off between recall and accuracy
- Ensuring operational stability while maintaining high detection coverage
- Evaluation is conducted using a combination of:
- Offline testing on prepared datasets

Selected near real-time evaluation scenarios, depending on the detection component

These evaluation methods ensure that the models are assessed under conditions relevant to practical DNS security monitoring environments.

6. ANTICIPATED BENEFITS

6.1. Benefits to Users

The proposed research is expected to benefit multiple user groups involved in DNS security monitoring and analysis, particularly within enterprise and organizational environments.

1. Security Analysts and SOC Teams

- Improved visibility into DNS-based threats through ML-driven anomaly detection capable of identifying diverse malicious behaviors.
- Enhanced detection of stealthy and low-frequency DNS threats, reducing the likelihood of missed anomalies during routine monitoring.
- Improved interpretability of detection outputs, supporting efficient investigation and informed decision-making.

2. Enterprise Security Teams and Network Administrators

- Support for centralized monitoring by integrating DNS anomaly detection outputs into existing enterprise security monitoring platforms.
- Increased adaptability of DNS threat analysis through the incorporation of contextual and environment-specific intelligence.
- Greater situational awareness of DNS-related risks across organizational networks.

3. Organizations and Operational Stakeholders

- Reduced exposure to prolonged DNS-based attacks through improved detection coverage and analytical depth.
- Enhanced confidence in DNS security monitoring without disrupting existing operational workflows.

6.2. Contribution to the Body of Knowledge

This research is expected to contribute to the cybersecurity domain by enhancing understanding of DNS-based threat detection within modern network environments. The study provides insights into how DNS-related security signals can be analyzed alongside intelligence-driven and analyst-informed context to address gaps in visibility, adaptability, and operational relevance.

- Contributes to cybersecurity research by examining DNS as a multi-dimensional attack surface and highlighting the challenges of detecting diverse DNS-based threats in enterprise and organizational settings.
- Advances knowledge on integrating anomaly detection with contextual and human intelligence to improve the interpretability and relevance of security monitoring outputs.
- Provides insight into the role of proactive and customizable threat intelligence in supporting security analysis and situational awareness across dynamic threat landscapes.
- Contributes experiential observations on training and evaluating ML-based DNS detection models, including considerations for balancing recall, interpretability, and operational scalability in cybersecurity monitoring environments.

7. SCOPE AND EXPECTED RESEARCH OUTCOMES

7.1. Scope of the Research

The scope of this research is focused on the detection and analysis of DNS-based threats using machine learning techniques, contextual threat intelligence, and analyst-provided inputs within modern network environments.

The research covers:

- Analysis of DNS traffic to identify anomalous and malicious behavior associated with multiple DNS threat categories, including DNS tunneling and data exfiltration, infrastructure abuse, DGA-based domains, and DNS-based command-and-control communication.
- Examination of DNS traffic under both traditional and encrypted DNS usage, including environments where DNS over HTTPS (DoH) is present, with emphasis on detection without reliance on payload inspection.
- Investigation of multiple machine learning approaches, including hybrid and fusion-based analytical techniques, to improve detection coverage and robustness across diverse DNS threat behaviors.
- Design and evaluation of ML-based detection approaches using features that can be extracted from live DNS traffic.
- Integration of proactive and customizable threat intelligence to support improved interpretation, contextualization, and prioritization of detection outputs.
- Support for centralized monitoring by enabling the integration of detection results into enterprise security monitoring platforms.

The research does not focus on automated mitigation, prevention, or response mechanisms. It is limited to detection, analysis, and contextual understanding of DNS-based security events within the defined research objectives.

7.2. Expected Deliverables and Outcomes

- A conceptual and experimental framework for ML-based DNS anomaly detection covering multiple DNS threat categories.

- Curated, cleaned, and processed DNS datasets suitable for training, validation, and evaluation of detection models.
- Feature sets and analytical insights relevant to real-time DNS anomaly detection.
- Evaluated ML-based detection models with emphasis on high recall, robustness, and operational relevance.
- A human-intelligence integration mechanism that allows analyst-provided context and environment-specific knowledge to enhance detection interpretation.
- Detection outputs that generate alerts with severity levels derived from prediction scores, accompanied by relevant contextual information such as source or principal identifiers, timestamps, and associated DNS attributes to support analyst-led incident analysis.
- Customizable threat intelligence dashboards that present detection results, contextual intelligence, and analytical summaries to support proactive investigation and situational awareness.
- Demonstration of centralized visibility through integration of detection outputs into enterprise security monitoring platforms.
- Documentation of experimental findings, observations, and research conclusions presented in the final project report.

8. RESEARCH CONSTRAINTS

This research is subject to several constraints that may influence the scope, methodology, and evaluation of the study.

1. Data Availability and Labeling

Access to complete, up-to-date, and accurately labeled DNS datasets covering a wide range of threat scenarios is limited. This may affect data balance and the ability of detection models to generalize across different types of DNS behavior.

2. Data Retention and Storage Constraints

The storage and retention of DNS traffic data and detection outputs are limited by storage capacity, privacy considerations, and the defined scope of the research. These limitations may reduce the ability to perform long-term analysis or study low-frequency threats over extended time periods.

3. Encrypted DNS Visibility

The growing use of encrypted DNS protocols reduces visibility into DNS query content. As a result, analysis is limited to observable metadata and traffic behavior rather than full inspection of query payloads.

4. Real-Time Processing Constraints

Designing detection approaches for near real-time analysis requires balancing model complexity, feature selection, and processing overhead. Only features that can be extracted efficiently from live DNS traffic can be used, which may limit the use of complex or long-duration analytical features.

5. Evaluation Environment Limitations

Testing and evaluation under large-scale enterprise traffic conditions may be limited by available infrastructure and resources. This can affect the assessment of system performance and scalability in real-world environments.

6. Threat Evolution and Concept Drift

DNS-based threats continue to evolve over time, which may reduce the effectiveness of trained models. Addressing long-term changes in threat behavior may require model updates beyond the scope of this research.

7. Integration and Deployment Constraints

Integration with external security monitoring platforms depends on system compatibility and available interfaces. These factors may limit the level of integration achievable within the project timeframe.

These constraints are considered throughout the research process to ensure that findings and conclusions remain realistic and appropriately scoped.

APPENDICES

Project Plan and Schedule

Initiation & Planning Phase	
• Define Research Problem	✓ Completed
• Literature Review	✓ Completed
• Project Proposal	✓ Completed
• Feasibility Study & Data Collection Planning	✓ Completed
• Tool & Technology Selection	✓ Completed
Design Phase	
• System Architecture	✓ Completed
• Feature Engineering	✓ Completed
• Dashboard Design	✓ Completed
Implementation Phase	
• Data Preprocessing	✓ Completed
• Model Training (Abuse, Tunneling, C2, TI)	✓ Completed
• Prototype UI	✓ Completed
Next Steps:	
• Rule Engine	⌘ Pending
• Alert Enrichment	⌘ Pending
• Inter Component Integration	⌘ Pending
• Integrating Logs into the SIEM (Wazuh)	⌘ Pending
Optimization Phase	
• Fine-tune Recall & Accuracy	⌘ Pending
• Real-time Performance Testing	⌘ Pending
Monitoring Phase	
• 2-Week Live Monitoring	⌘ Pending
• Accuracy and Recall Refinement	⌘ Pending
Closing Phase	
• Writing Research Paper	⌘ Pending
• Final Progress Presentation (PP2)	⌘ Pending

Figure 1: Project Plan and Schedule

Current System Architectures

DNS Tunneling and Data Exfiltration Detection

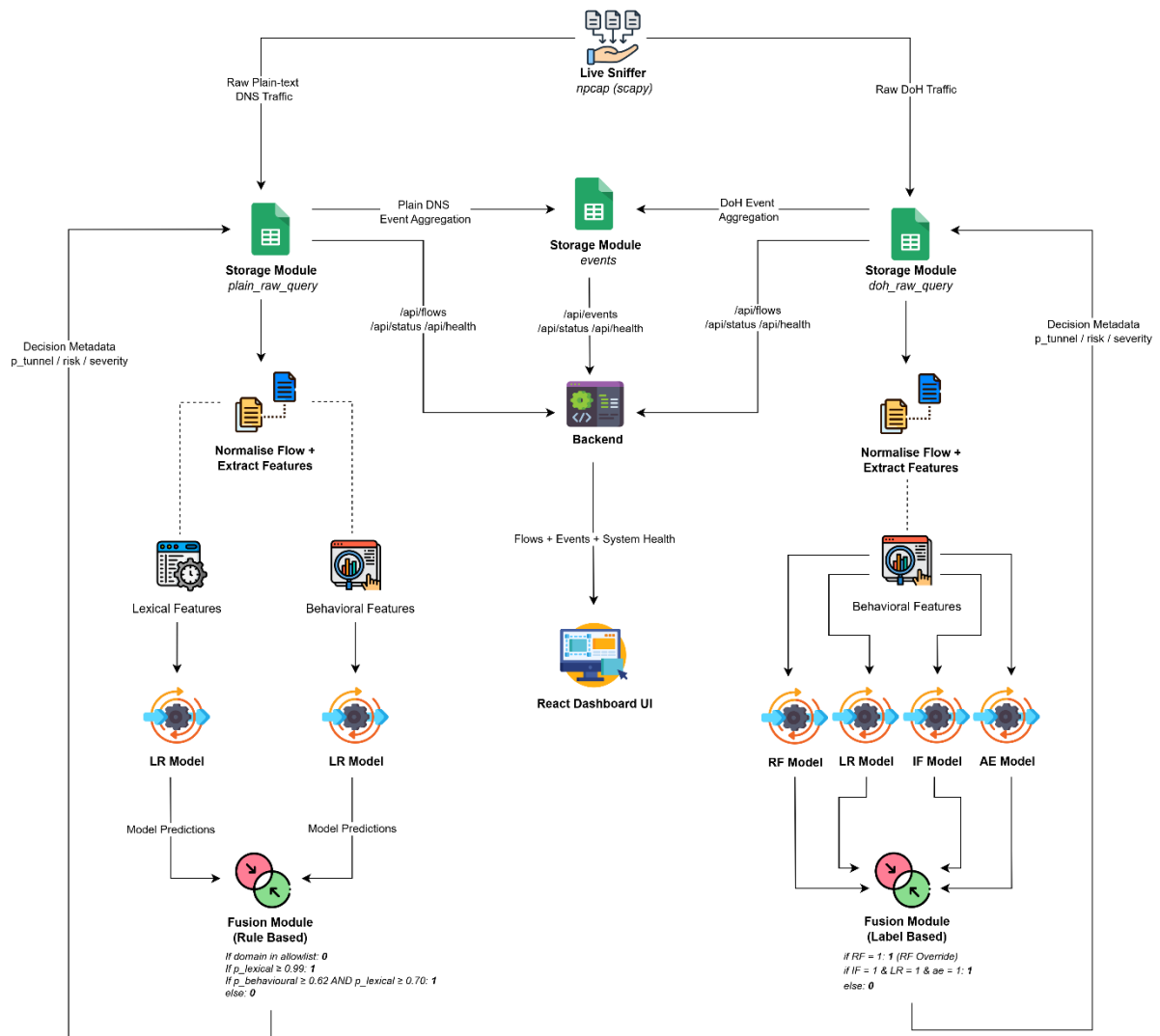


Figure 2: DNS Tunneling and Data Exfiltration Detection System Architecture

DGA-Based DNS and C2 Communication Detection

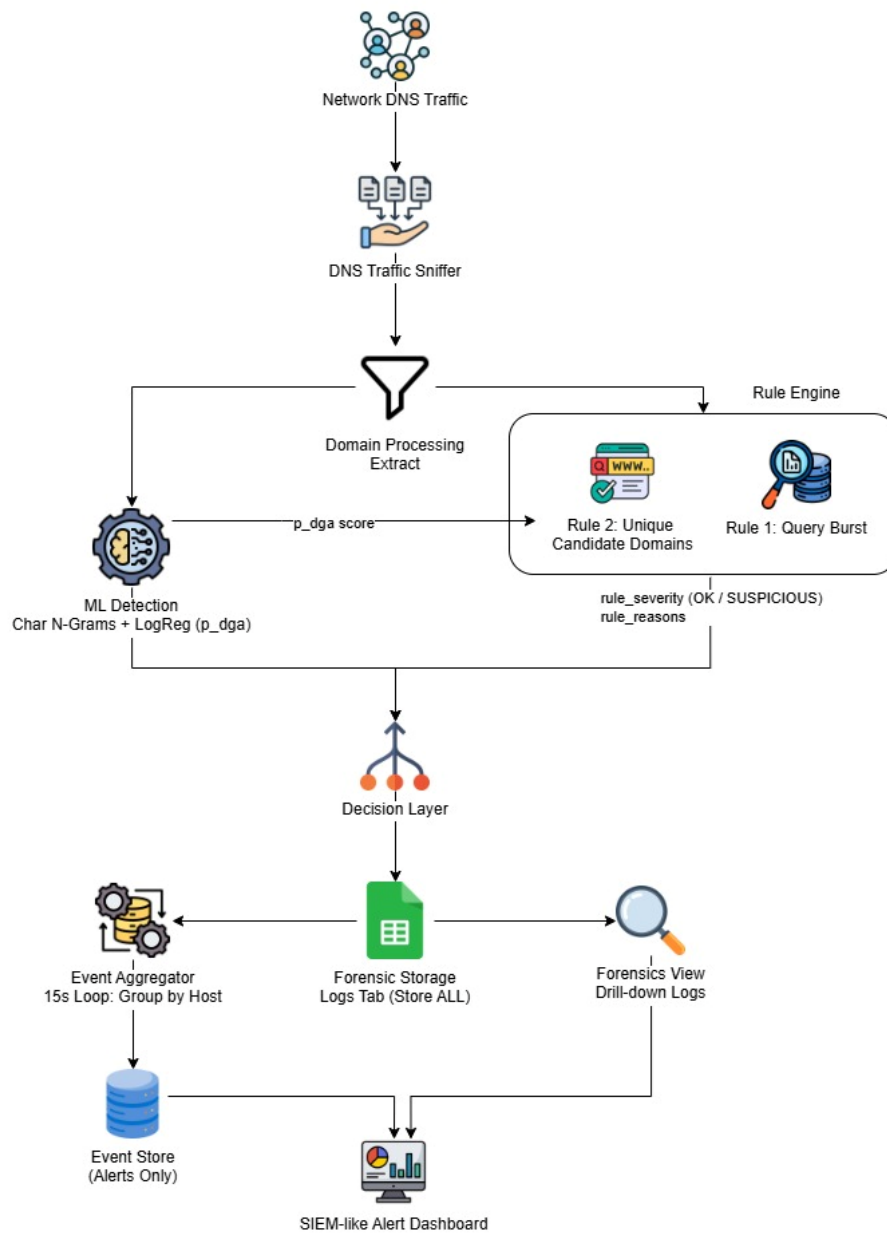


Figure 3: DGA-Based DNS and C2 Communication Detection System Architecture

DNS Abuse and Infrastructure Attack Detection

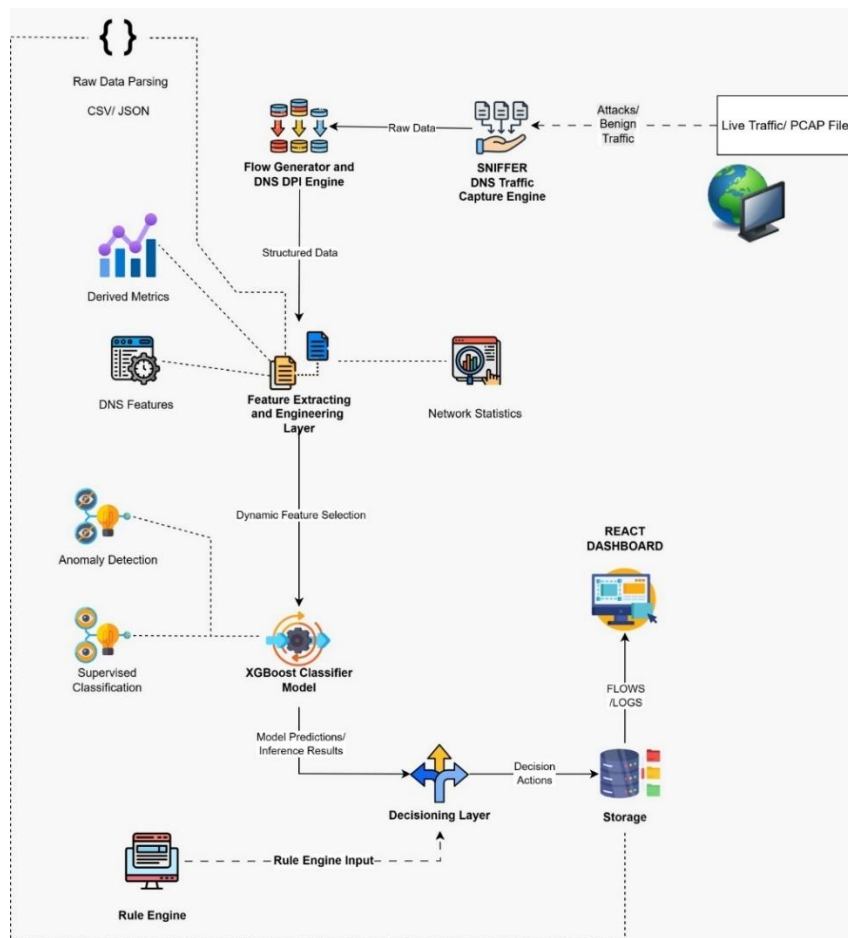


Figure 4: DNS Abuse and Infrastructure Attack Detection System Architecture

Threat Intelligence Correlation and Scoring

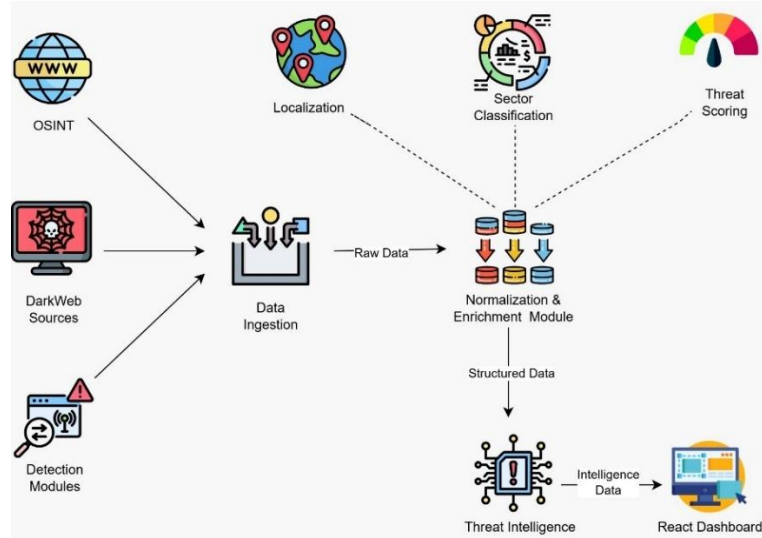


Figure 5: Threat Intelligence Correlation and Scoring System Architecture

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